88 *PT* AFM collinear spin layer groups

Néel directions, SOC magnetic layer groups and layer-resolved intrinsic nonlinear Hall constraints

Coverage. All 88 entries of Table S5, in the original order, expanded into **625 direction branches**: 1 group has 1 branch, 1 has 2, 20 have 3, 42 have 7, 4 have 8, 16 have 11, and 4 have 15. Directions related by a parent symmetry are retained in the fixed source coordinate setting so that the displayed tensor indices stay meaningful. These 625 rows are direction strata, not 625 distinct MLG types.

Source. J.-X. Du, S. Zeng and Y.-J. Zhao, *Stacking theory for bilayer two-dimensional magnets*, Phys. Rev. B **113**, 214455 (2026), DOI: 10.1103/g471-xrkl; Supplemental Material, Table S5, pp. 64–84. The source supplies the cSLGs and operations. The Néel-direction branches, SOC MLGs and response constraints in this report are calculated results.

Direction notation. $\mathrm{AX}_{[uvw]}$ denotes a special axis; $\mathrm{PL}_{(hkl)}$ denotes a direction in a special plane excluding all axes lying in that plane; GEN denotes a direction outside every listed axis and plane. Only symmetry-distinct stabilizer strata of a given group are introduced; for example, group 2.3.6 has a single GEN row covering every direction; group 66.3.466 needs only $\mathrm{AX}_{[001]}$ and GEN.

Coordinates. Tensor components are Cartesian, with z normal to the layer. For rectangular and tetragonal groups, $x \parallel \mathbf{a}$, $y \parallel \mathbf{b}$. For hexagonal groups, $\mathbf{a} = (1, 0, 0)$ and $\mathbf{b} = (-1/2, \sqrt{3}/2, 0)$, while Cartesian $y \parallel \hat{z} \times \mathbf{a}$. Miller indices refer to these crystallographic bases. Thus [010] is not the Cartesian y direction in the hexagonal setting. Hexagonal in-plane operations are displayed with explicit crystallographic directions; $M_{[uvw]}$ has its Cartesian normal along $[uvw]$. The source triplets $\{x, y, xy\}$ and $\{1, 2, 3\}$ correspond as sets to $\{[100], [010], [110]\}$ and $\{[1\bar{1}0], [120], [210]\}$. No individual identification of the source's numbered subscripts is needed.

Operation columns. Translation parts are omitted and repeated point operations are merged. All distinct nontrivial-part SLG point operations are shown on separate lines; the common spin-only factor is implicit. Every MLG row explicitly lists its point operations on one line. Full MLG symbols and numbers are nevertheless identified from the full affine operations in the source, then verified by a basis/origin change preserving the layer normal. MLG symbols use their standard settings; the explicitly listed operations and tensor indices remain in the source coordinate setting. An affine witness for every row is included in the data package. Omitting translations from the displayed columns does not redefine these group identifiers.

Tensor convention. Set $A^\ell = \chi^\ell_{xyy}$ and $B^\ell = \chi^\ell_{yxx}$ for $\ell = t, b$, with $\chi_{abc} = -\chi_{bac}$ and in-plane indices. Then $\chi_{xyx} = -B$, $\chi_{yxy} = -A$, and all other in-plane entries vanish. The two-component vector $\mathbf{v}^\ell = (A^\ell, B^\ell)$ transforms as $\mathbf{v}^{g\ell} = q_g R_\parallel \mathbf{v}^\ell$. This statement is specific to the $\mathcal{T}$-odd intrinsic Hall part.

Mark	Constraint on the displayed component	Meaning
$+$	$\chi^t = \chi^b$, allowed	Equal layer contributions
$-$	$\chi^t = -\chi^b$, allowed	Hidden, layer compensated
0	$\chi^t = \chi^b = 0$	Forbidden in each layer
v	No fixed proportional relation between the same component in the two layers	See coupled constraints when starred

A star on a direction indicates additional relations between A and B, listed completely in Appendix A (104 branches). In these rows, v must not be read as absence of all symmetry relations: cross-component relations may remain. All allowed values can accidentally vanish microscopically. Every branch retains $\mathcal{PT}$, which exchanges the two layers and enforces $A^t = A^b$ and $B^t = B^b$. Only $+$ and 0 occur in this class. Other symmetries may forbid a component or relate A and B; Appendix B gives the general PT argument.

Assumptions and calculation. Fixed structure and collinear order; two physical layers at opposite sides of $z = 0$. The SOC subgroup satisfies $q_g \det(R_g) R_g \hat{\mathbf{n}} = s_g \hat{\mathbf{n}}$, with $q_g = +1$ for g and $q_g = -1$ for $g\mathcal{T}$. Operations with $R_{zz} = -1$ exchange the two layers. Branches are constructed from all real ± 1 eigenspaces and their intersections. These are symmetry-allowed constrained orientations, not predictions of magnetic easy axes or stability.

Source audit (S1). In 39.3.258, 39.6.261 and 39.7.262, operation 08 on p. 67 assigns translation $(1/2, 0, 0)$ to M_x. Multiplying the listed inversion operation 05 by C_{2x} operation 04 instead gives $(0, -1/2, 0) \equiv (0, 1/2, 0)$. We use $(0, 1/2, 0)$ in all three entries; their spin labels are unchanged. All 88 corrected operation lists close and their spin labels agree with the printed cSLG symbols. Original strings are retained in the raw data. These are consistency repairs, not author-issued errata; they affect full affine identification but not the displayed point operations or tensor constraints.

Validation. All 625 MLG assignments are checked against complete affine operations. Each allowed response space is independently checked using the full third-rank Hall tensor, and direction coverage is sampled on every axis and plane as well as generic orientations.

Complete table, in Table S5 order

First three columns are merged across all direction branches of one cSLG. Asterisks refer to Appendix A; S1 refers to the source audit on the first page.

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
01 2.3.6 $p^{-1}\bar{1}$ SI p. 64	PT AFM	$\{1\|E\}$ $\{-1\|\mathcal{P}\}$	GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
02 6.3.23 $p^{1}1^{1}1^{-1}2/^{1}m$ SI p. 64	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p112'/m$ 6.3.23	$\{E,\, C_{2z}\mathcal{T},\, \mathcal{PT},\, M_z\}$	+	+
			$\mathrm{PL}_{(001)}$	$p112/m'$ 6.4.24	$\{E,\, C_{2z},\, \mathcal{PT},\, M_z\mathcal{T}\}$	0	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
03 6.4.24 $p^{1}1^{1}1^{1}2/^{-1}m$ SI p. 64	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p112/m'$ 6.4.24	$\{E,\, C_{2z},\, \mathcal{PT},\, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E,\, C_{2z}\mathcal{T},\, \mathcal{PT},\, M_z\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
04 7.3.30 $p^{1}1^{1}1^{-1}2/^{1}a$ SI p. 64	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p112'/a$ 7.3.30	$\{E,\, C_{2z}\mathcal{T},\, \mathcal{PT},\, M_z\}$	+	+
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E,\, C_{2z},\, \mathcal{PT},\, M_z\mathcal{T}\}$	0	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
05 7.4.31 $p^{1}1^{1}1^{1}2/^{-1}a$ SI p. 64	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[001]}$	$p112/a'$ 7.4.31	$\{E,\, C_{2z},\, \mathcal{PT},\, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E,\, C_{2z}\mathcal{T},\, \mathcal{PT},\, M_z\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
06 14.3.68 $p^{1}2/^{-1}m^{1}1^{1}1$ SI p. 64	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2/m'11$ 14.3.68	$\{E,\, C_{2x},\, \mathcal{PT},\, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/m11$ 14.4.69	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{PT},\, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
07 14.4.69 $p^{-1}2/^{1}m^{1}1^{1}1$ SI p. 64	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2'/m11$ 14.4.69	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{PT},\, M_x\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/m'11$ 14.3.68	$\{E,\, C_{2x},\, \mathcal{PT},\, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
08 15.3.78 $p^{1}2_1/^{-1}m^{1}1^{1}1$ SI p. 64	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2_1/m'11$ 15.3.78	$\{E,\, C_{2x},\, \mathcal{PT},\, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1'/m11$ 15.5.80	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{PT},\, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+
09 15.5.80 $p^{-1}2_1/^{1}m^{1}1^{1}1$ SI p. 64	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$p2_1'/m11$ 15.5.80	$\{E,\, C_{2x}\mathcal{T},\, \mathcal{PT},\, M_x\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E,\, C_{2x},\, \mathcal{PT},\, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E,\, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
10 16.3.85 $p^12/^{-1}b^111$ SI p. 64	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$PL_{(100)}$	$p2'/b11$ 16.5.87	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
11 16.5.87 $p^{-1}2/^1b^111$ SI p. 64	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$p2'/b11$ 16.5.87	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$PL_{(100)}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
12 17.3.92 $p^12_1/^{-1}b^111$ SI p. 65	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$PL_{(100)}$	$p2_1'/b11$ 17.5.94	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
13 17.5.94 $p^{-1}2_1/^1b^111$ SI p. 65	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$p2_1'/b11$ 17.5.94	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$PL_{(100)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
14 18.3.97 $c^12/^{-1}m^111$ SI p. 65	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$c2/m'11$ 18.3.97	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$PL_{(100)}$	$c2'/m11$ 18.5.99	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
15 18.5.99 $c^{-1}2/^1m^111$ SI p. 65	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$	$AX_{[100]}$	$c2'/m11$ 18.5.99	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$PL_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
16 37.3.232 $p^{-1}m^{-1}m^{-1}m$ SI p. 65	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$AX_{[010]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$AX_{[001]}$	$pmmm'$ 37.6.235	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$PL_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$PL_{(010)}$	$p2'/m11$ 14.4.69	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$PL_{(100)}$	$p2'/m11$ 14.4.69	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
17 37.6.235 $p^1m^1m^{-1}m$ SI p. 65	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$AX_{[010]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$AX_{[001]}$	$pm'm'm'$ 37.3.232	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$PL_{(010)}$	$p2/m'11$ 14.3.68	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$PL_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
18 37.7.236 $p^1m^{-1}m^1m$ SI p. 65	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$pmmm'$ 37.6.235	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$AX_{[010]}$	$pm'm'm'$ 37.3.232	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$PL_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$PL_{(010)}$	$p2'/m11$ 14.4.69	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$PL_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
19 38.3.245 $p^{-1}m^{-1}a^{-1}a$ SI p. 66	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$	$AX_{[100]}$	$pm'aa$ 38.7.249	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z, M_y\}$	+	0
			$AX_{[010]}$	$pma'a$ 38.9.251	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_x, M_z, M_y\mathcal{T}\}$	0	+
			$AX_{[001]}$	$pmaa'$ 38.8.250	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$PL_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$PL_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$PL_{(100)}$	$p2'/m11$ 14.4.69	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
20 38.7.249 $p^{-1}m^1a^1a$ SI p. 66	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{1\|M_y\}$	$AX_{[100]}$	$pm'a'a'$ 38.3.245	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$AX_{[010]}$	$pmaa'$ 38.8.250	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$AX_{[001]}$	$pma'a$ 38.9.251	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_x, M_z, M_y\mathcal{T}\}$	0	+
			$PL_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$PL_{(010)}$	$p2/b'11$ 16.3.85	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$PL_{(100)}$	$p2'/m11$ 14.4.69	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
21 38.8.250 $p^1m^1a^{-1}a$ SI p. 66	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pma'a$ 38.9.251	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{PT}, M_x, M_z, M_y\mathcal{T}\}$	0	+
			$\mathrm{AX}_{[010]}$	$pm'aa$ 38.7.249	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}, M_z, M_y\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'a'a'$ 38.3.245	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{PT}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/b'11$ 16.3.85	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
22 38.9.251 $p^1m^{-1}a^1a$ SI p. 66	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pmaa'$ 38.8.250	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{PT}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'a'a'$ 38.3.245	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{PT}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pm'aa$ 38.7.249	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{PT}, M_x\mathcal{T}, M_z, M_y\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
23 (S1) 39.3.258 $p^{-1}b^{-1}a^{-1}n$ SI p. 66	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pban'$ 39.6.261	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/b11$ 16.5.87	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
24 (S1) 39.6.261 $p^1b^1a^{-1}n$ SI p. 67	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pb'a'n'$ 39.3.258	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{PT}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/b'11$ 16.3.85	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
25 (S1) 39.7.262 $p^1b^{-1}a^1n$ SI p. 67	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pban'$ 39.6.261	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pb'a'n'$ 39.3.258	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
26 40.3.265 $p^{-1}m^{-1}a^{-1}m$ SI p. 67	PT AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'am$ 40.9.271	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pma'm$ 40.7.269	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pmam'$ 40.8.270	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
27 40.7.269 $p^1m^{-1}a^1m$ SI p. 67	PT AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmam'$ 40.8.270	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'a'm'$ 40.3.265	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pm'am$ 40.9.271	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
28 40.8.270 $p^1m^1a^{-1}m$ SI p. 67	PT AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pma'm$ 40.7.269	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pm'am$ 40.9.271	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'a'm'$ 40.3.265	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/b'11$ 16.3.85	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
29 40.9.271 $p^{-1}m^1a^1m$ SI p. 67	PT AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'a'm'$ 40.3.265	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pmam'$ 40.8.270	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$pma'm$ 40.7.269	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z, M_x\}$	0	+
			$\mathrm{PL}_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2/b'11$ 16.3.85	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
30 41.3.278 $p^{-1}m^{-1}m^{-1}a$ SI p. 68	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'ma$ 41.9.284	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pmm'a$ 41.8.283	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pmma'$ 41.7.282	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'/m11$ 14.4.69	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
31 41.7.282 $p^1m^1m^{-1}a$ SI p. 68	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'a$ 41.8.283	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pm'ma$ 41.9.284	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'm'a'$ 41.3.278	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/m'11$ 14.3.68	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
32 41.8.283 $p^1m^{-1}m^1a$ SI p. 68	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmma'$ 41.7.282	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'm'a'$ 41.3.278	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pm'ma$ 41.9.284	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'/m11$ 14.4.69	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
33 41.9.284 $p^{-1}m^1m^1a$ SI p. 68	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'm'a'$ 41.3.278	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pmma'$ 41.7.282	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$pmm'a$ 41.8.283	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2/m'11$ 14.3.68	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
34 42.3.291 $p^{-1}m^{-1}a^{-1}n$ SI p. 68	PT AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'an$ 42.9.297	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pma'n$ 42.7.295	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pman'$ 42.8.296	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/m11$ 14.4.69	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
35 42.7.295 $p^1m^{-1}a^1n$ SI p. 69	PT AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pman'$ 42.8.296	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'a'n'$ 42.3.291	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pm'an$ 42.9.297	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
36 42.8.296 $p^1m^1a^{-1}n$ SI p. 69	PT AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_z\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pma'n$ 42.7.295	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pm'an$ 42.9.297	$\{E, C_{2y}\mathcal{T}, C_{2z}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_y, M_z, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'a'n'$ 42.3.291	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
37 42.9.297 $p^{-1}m^1a^1n$ SI p. 69	PT AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_z\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pm'a'n'$ 42.3.291	$\{E, C_{2y}, C_{2z}, C_{2x}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pman'$ 42.8.296	$\{E, C_{2y}\mathcal{T}, C_{2z}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_z\mathcal{T}, M_x\}$	0	0
			$\mathrm{AX}_{[001]}$	$pma'n$ 42.7.295	$\{E, C_{2y}, C_{2z}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_z, M_x\}$	0	+
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2'/m11$ 14.4.69	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
38 43.3.300 $p^{-1}b^{-1}a^{-1}a$ SI p. 69	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pb'aa$ 43.7.304	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z, M_y\}$	+	0
			$\mathrm{AX}_{[010]}$	$pba'a$ 43.9.306	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_x, M_z, M_y\mathcal{T}\}$	0	+
			$\mathrm{AX}_{[001]}$	$pbaa'$ 43.8.305	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/b11$ 16.5.87	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
39 43.7.304 $p^{-1}b^1a^1a$ SI p. 69	PT AFM	$\{1\|E\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pb'a'a'$ 43.3.300	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pbaa'$ 43.8.305	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$\mathrm{AX}_{[001]}$	$pba'a$ 43.9.306	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_x, M_z, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2'/b11$ 16.5.87	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
40 43.8.305 $p^1b^1a^{-1}a$ SI p. 69	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pba'a$ 43.9.306	$\{E, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, C_{2y}, \mathcal{P}\mathcal{T}, M_x, M_z, M_y\mathcal{T}\}$	0	+
			$\mathrm{AX}_{[010]}$	$pb'aa$ 43.7.304	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z, M_y\}$	+	0
			$\mathrm{AX}_{[001]}$	$pb'a'a'$ 43.3.300	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
41 43.9.306 $p^1b^{-1}a^1a$ SI p. 70	PT AFM	$\{1\|E\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_x\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$	$\mathrm{AX}_{[100]}$	$pbaa'$ 43.8.305	$\{E, C_{2x}\mathcal{T}, C_{2z}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x, M_z\mathcal{T}, M_y\}$	0	0
			$\mathrm{AX}_{[010]}$	$pb'a'a'$ 43.3.300	$\{E, C_{2x}, C_{2z}, C_{2y}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pb'aa$ 43.7.304	$\{E, C_{2x}, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}, M_z, M_y\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
42 44.3.309 $p^{-1}b^{-1}a^{-1}m$ SI p. 70	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pbam'$ 44.6.312	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
43 44.6.312 $p^1b^1a^{-1}m$ SI p. 70	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pb'a'm'$ 44.3.309	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
44 44.7.313 $p^1b^{-1}a^1m$ SI p. 70	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pbam'$ 44.6.312	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pb'a'm'$ 44.3.309	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
45 45.3.316 $p^{-1}b^{-1}m^{-1}a$ SI p. 70	PT AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb'ma$ 45.8.321	$\{E, C_{2y}\mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_x\mathcal{T}, M_z\}$	+	0
			$\mathrm{AX}_{[010]}$	$pbm'a$ 45.7.320	$\{E, C_{2y}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_x, M_z\}$	0	+
			$\mathrm{AX}_{[001]}$	$pbma'$ 45.9.322	$\{E, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_y, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
46 45.7.320 $p^{1}b^{-1}m^{1}a$ SI p. 71	PT AFM	$\{1\|E\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pbma'$ 45.9.322	$\{E, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_y, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pb'm'a'$ 45.3.316	$\{E, C_{2y}, C_{2x}, C_{2z}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pb'ma$ 45.8.321	$\{E, C_{2y}\mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_x\mathcal{T}, M_z\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
47 45.8.321 $p^{-1}b^{1}m^{1}a$ SI p. 71	PT AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pb'm'a'$ 45.3.316	$\{E, C_{2y}, C_{2x}, C_{2z}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$pbma'$ 45.9.322	$\{E, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2z}, \mathcal{P}\mathcal{T}, M_y, M_x, M_z\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pbm'a$ 45.7.320	$\{E, C_{2y}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_x, M_z\}$	0	+
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2'_1/b11$ 17.5.94	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
48 45.9.322 $p^{1}b^{1}m^{-1}a$ SI p. 71	PT AFM	$\{1\|E\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2z}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_z\}$	$\mathrm{AX}_{[100]}$	$pbm'a$ 45.7.320	$\{E, C_{2y}, C_{2x}\mathcal{T}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_x, M_z\}$	0	+
			$\mathrm{AX}_{[010]}$	$pb'ma$ 45.8.321	$\{E, C_{2y}\mathcal{T}, C_{2x}, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y, M_x\mathcal{T}, M_z\}$	+	0
			$\mathrm{AX}_{[001]}$	$pb'm'a'$ 45.3.316	$\{E, C_{2y}, C_{2x}, C_{2z}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
49 46.3.325 $p^{-1}m^{-1}m^{-1}n$ SI p. 71	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$pmmn'$ 46.6.328	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
50 46.6.328 $p^1m^1m^{-1}n$ SI p. 71	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[001]}$	$pm'm'n'$ 46.3.325	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{PT}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
51 46.7.329 $p^1m^{-1}m^1n$ SI p. 71	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$\mathrm{AX}_{[100]}$	$pmmn'$ 46.6.328	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{AX}_{[010]}$	$pm'm'n'$ 46.3.325	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{PT}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$p2'_1/m11$ 15.5.80	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
52 47.3.332 $c^{-1}m^{-1}m^{-1}m$ SI p. 72	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$\mathrm{AX}_{[100]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[001]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$c2'/m11$ 18.5.99	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$c2'/m11$ 18.5.99	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
53 47.6.335 $c^1m^1m^{-1}m$ SI p. 72	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$AX_{[010]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$AX_{[001]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$PL_{(010)}$	$c2/m'11$ 18.3.97	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$PL_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
54 47.7.336 $c^1m^{-1}m^1m$ SI p. 72	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$AX_{[010]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$PL_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$PL_{(010)}$	$c2'/m11$ 18.5.99	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$PL_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
55 48.3.345 $c^{-1}m^{-1}m^{-1}e$ SI p. 73	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$	$AX_{[100]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$AX_{[010]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$AX_{[001]}$	$cmme'$ 48.6.348	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$PL_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$PL_{(010)}$	$c2'/m11$ 18.5.99	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$PL_{(100)}$	$c2'/m11$ 18.5.99	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
56 48.6.348 $c^1m^1m^{-1}e$ SI p. 73	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$AX_{[010]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$AX_{[001]}$	$cm'm'e'$ 48.3.345	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$PL_{(010)}$	$c2/m'11$ 18.3.97	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$PL_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
57 48.7.349 $c^1m^{-1}m^1e$ SI p. 73	PT AFM	$\{1\|E\}$ $\{-1\|C_{2z}\}$ $\{1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|\mathcal{P}\}$ $\{1\|M_z\}$ $\{-1\|M_y\}$ $\{1\|M_x\}$	$AX_{[100]}$	$cmme'$ 48.6.348	$\{E, C_{2z}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y, M_x\}$	0	0
			$AX_{[010]}$	$cm'm'e'$ 48.3.345	$\{E, C_{2z}, C_{2y}, C_{2x}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}\}$	0	0
			$AX_{[001]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$PL_{(001)}$	$p112/a'$ 7.4.31	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$PL_{(010)}$	$c2'/m11$ 18.5.99	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$PL_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
58 51.3.364 $p^14/^{-1}m$ SI p. 74	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$	$AX_{[001]}$	$p4/m'$ 51.3.364	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
59 51.4.365 $p^{-1}4/^{-1}m$ SI p. 74	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$	$AX_{[001]}$	$p4'/m'$ 51.4.365	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}\}$	0	0
			$PL_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
60 52.3.371 $p^14/^{-1}n$ SI p. 74	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$	$AX_{[001]}$	$p4/n'$ 52.3.371	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}\}$	0	0
			$PL_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
61 52.4.372 $p^{-1}4/^{-1}n$ SI p. 74	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p4'/n'$ 52.4.372	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
62 61.3.424 $p^{1}4/^{-1}m^{-1}m^{-1}m$ SI p. 75	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^{*}_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/m'mm$ 61.5.426	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'/m11$ 14.4.69	$\{E, C_{2y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/m11$ 14.4.69	$\{E, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_x\}$	0	+
			$\mathrm{PL}^{*}_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{xy}\}$	+	+
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
63 61.5.426 $p^{1}4/^{-1}m^{1}m^{1}m$ SI p. 75	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{P}\mathcal{T}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^{*}_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/m'm'm'$ 61.3.424	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/m'11$ 14.3.68	$\{E, C_{2y}, \mathcal{P}\mathcal{T}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{P}\mathcal{T}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^{*}_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{P}\mathcal{T}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{P}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
64 61.6.427 $p^{-1}4/^{-1}m^{-1}m^{1}m$ SI p. 75	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^{*}_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/m'mm'$ 61.7.428	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'/m11$ 14.4.69	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/m11$ 14.4.69	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$\mathrm{PL}^{*}_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
65 61.7.428 $p^{-1}4/^{-1}m^{1}m^{-1}m$ SI p. 76	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pmm'm$ 37.7.236	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^{*}_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/m'm'm$ 61.6.427	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/m'11$ 14.3.68	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/m'11$ 14.3.68	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^{*}_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{PT}, M_{xy}\}$	+	+
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
66 62.3.437 $p^14/^{-1}n^{-1}b^{-1}m$ SI p. 76	PT AFM	$\{1\|E\}$, $\{1\|C_{2z}\}$, $\{1\|C_{4z}\}$, $\{1\|C_{4z}^{-1}\}$, $\{1\|C_{2y}\}$, $\{1\|C_{2x}\}$, $\{1\|C_{2xy}\}$, $\{1\|C_{2\bar{x}y}\}$, $\{-1\|\mathcal{P}\}$, $\{-1\|M_z\}$, $\{-1\|\bar{C}_{4z}\}$, $\{-1\|\bar{C}_{4z}^{-1}\}$, $\{-1\|M_y\}$, $\{-1\|M_x\}$, $\{-1\|M_{xy}\}$, $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/n'bm$ 62.5.439	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/b11$ 16.5.87	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{PT}, M_{xy}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
67 62.5.439 $p^14/^{-1}n^1b^1m$ SI p. 76	PT AFM	$\{1\|E\}$, $\{1\|C_{2z}\}$, $\{1\|C_{4z}\}$, $\{1\|C_{4z}^{-1}\}$, $\{-1\|C_{2y}\}$, $\{-1\|C_{2x}\}$, $\{-1\|C_{2xy}\}$, $\{-1\|C_{2\bar{x}y}\}$, $\{-1\|\mathcal{P}\}$, $\{-1\|M_z\}$, $\{-1\|\bar{C}_{4z}\}$, $\{-1\|\bar{C}_{4z}^{-1}\}$, $\{1\|M_y\}$, $\{1\|M_x\}$, $\{1\|M_{xy}\}$, $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/n'b'm'$ 62.3.437	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/b'11$ 16.3.85	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
68 62.6.440 $p^{-1}4/^{-1}n^{-1}b^{1}m$ SI p. 77	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/n'bm'$ 62.7.441	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2'/b11$ 16.5.87	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2'/b11$ 16.5.87	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
69 62.7.441 $p^{-1}4/^{-1}n^{1}b^{-1}m$ SI p. 77	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pba'n$ 39.7.262	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/n'b'm$ 62.6.440	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2/b'11$ 16.3.85	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2/b'11$ 16.3.85	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{PT}, M_{xy}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
70 63.3.446 $p^14/^{-1}m^{-1}b^{-1}m$ SI p. 77	PT AFM	$\{1\|E\}$, $\{1\|C_{2z}\}$, $\{1\|C_{4z}\}$, $\{1\|C_{4z}^{-1}\}$, $\{1\|C_{2y}\}$, $\{1\|C_{2x}\}$, $\{1\|C_{2xy}\}$, $\{1\|C_{2\bar{x}y}\}$, $\{-1\|\mathcal{P}\}$, $\{-1\|M_z\}$, $\{-1\|\bar{C}_{4z}\}$, $\{-1\|\bar{C}_{4z}^{-1}\}$, $\{-1\|M_y\}$, $\{-1\|M_x\}$, $\{-1\|M_{xy}\}$, $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^*_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/m'bm$ 63.5.448	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1'/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1'/b11$ 17.5.94	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{PT}, M_{xy}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
71 63.5.448 $p^14/^{-1}m^1b^1m$ SI p. 78	PT AFM	$\{1\|E\}$, $\{1\|C_{2z}\}$, $\{1\|C_{4z}\}$, $\{1\|C_{4z}^{-1}\}$, $\{-1\|C_{2y}\}$, $\{-1\|C_{2x}\}$, $\{-1\|C_{2xy}\}$, $\{-1\|C_{2\bar{x}y}\}$, $\{-1\|\mathcal{P}\}$, $\{-1\|M_z\}$, $\{-1\|\bar{C}_{4z}\}$, $\{-1\|\bar{C}_{4z}^{-1}\}$, $\{1\|M_y\}$, $\{1\|M_x\}$, $\{1\|M_{xy}\}$, $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/m'b'm'$ 63.3.446	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
72 63.6.449 $p^{-1}4/^{-1}m^{-1}b^{1}m$ SI p. 78	PT AFM	$\{1\|E\}$, $\{1\|C_{2z}\}$, $\{-1\|C_{4z}\}$, $\{-1\|C_{4z}^{-1}\}$, $\{1\|C_{2y}\}$, $\{1\|C_{2x}\}$, $\{-1\|C_{2xy}\}$, $\{-1\|C_{2\bar{x}y}\}$, $\{-1\|\mathcal{P}\}$, $\{-1\|M_z\}$, $\{1\|\bar{C}_{4z}\}$, $\{1\|\bar{C}_{4z}^{-1}\}$, $\{-1\|M_y\}$, $\{-1\|M_x\}$, $\{1\|M_{xy}\}$, $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^*_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/m'bm'$ 63.7.450	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1'/b11$ 17.5.94	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1'/b11$ 17.5.94	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
73 63.7.450 $p^{-1}4/^{-1}m^{1}b^{-1}m$ SI p. 79	PT AFM	$\{1\|E\}$, $\{1\|C_{2z}\}$, $\{-1\|C_{4z}\}$, $\{-1\|C_{4z}^{-1}\}$, $\{-1\|C_{2y}\}$, $\{-1\|C_{2x}\}$, $\{1\|C_{2xy}\}$, $\{1\|C_{2\bar{x}y}\}$, $\{-1\|\mathcal{P}\}$, $\{-1\|M_z\}$, $\{1\|\bar{C}_{4z}\}$, $\{1\|\bar{C}_{4z}^{-1}\}$, $\{1\|M_y\}$, $\{1\|M_x\}$, $\{-1\|M_{xy}\}$, $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pba'm$ 44.7.313	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/m'b'm$ 63.6.449	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/b'11$ 17.3.92	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{PT}, M_{xy}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
74 64.3.455 $p^14/^{-1}n^{-1}m^{-1}m$ SI p. 79	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/n'mm$ 64.5.457	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y, M_x, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1'/m11$ 15.5.80	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1'/m11$ 15.5.80	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{PT}, M_{xy}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
75 64.5.457 $p^14/^{-1}n^1m^1m$ SI p. 79	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{1\|C_{4z}\}$ $\{1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{4z}\}$ $\{-1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4/n'm'm'$ 64.3.455	$\{E, C_{2z}, C_{4z}, C_{4z}^{-1}, C_{2y}, C_{2x}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}\mathcal{T}, \bar{C}_{4z}^{-1}\mathcal{T}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
76 64.6.458 $p^{-1}4/^{-1}n^{-1}m^{1}m$ SI p. 80	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{1\|C_{2y}\}$ $\{1\|C_{2x}\}$ $\{-1\|C_{2xy}\}$ $\{-1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{-1\|M_y\}$ $\{-1\|M_x\}$ $\{1\|M_{xy}\}$ $\{1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}_{[010]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/n'mm'$ 64.7.459	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y, M_x, M_{xy}\mathcal{T}, M_{\bar{x}y}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1'/m11$ 15.5.80	$\{E, C_{2y}\mathcal{T}, \mathcal{PT}, M_y\}$	+	0
			$\mathrm{PL}_{(100)}$	$p2_1'/m11$ 15.5.80	$\{E, C_{2x}\mathcal{T}, \mathcal{PT}, M_x\}$	0	+
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2xy}, \mathcal{PT}, M_{xy}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2\bar{x}y}, \mathcal{PT}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
77 64.7.459 $p^{-1}4/^{-1}n^{1}m^{-1}m$ SI p. 80	PT AFM	$\{1\|E\}$ $\{1\|C_{2z}\}$ $\{-1\|C_{4z}\}$ $\{-1\|C_{4z}^{-1}\}$ $\{-1\|C_{2y}\}$ $\{-1\|C_{2x}\}$ $\{1\|C_{2xy}\}$ $\{1\|C_{2\bar{x}y}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|M_z\}$ $\{1\|\bar{C}_{4z}\}$ $\{1\|\bar{C}_{4z}^{-1}\}$ $\{1\|M_y\}$ $\{1\|M_x\}$ $\{-1\|M_{xy}\}$ $\{-1\|M_{\bar{x}y}\}$	$\mathrm{AX}_{[100]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}, C_{2x}\mathcal{T}, \mathcal{PT}, M_z, M_y\mathcal{T}, M_x\}$	0	+
			$\mathrm{AX}_{[010]}$	$pmm'n$ 46.7.329	$\{E, C_{2z}\mathcal{T}, C_{2y}\mathcal{T}, C_{2x}, \mathcal{PT}, M_z, M_y, M_x\mathcal{T}\}$	+	0
			$\mathrm{AX}^*_{[110]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z, M_{xy}\mathcal{T}, M_{\bar{x}y}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'e$ 48.7.349	$\{E, C_{2z}\mathcal{T}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}, \mathcal{PT}, M_z, M_{xy}, M_{\bar{x}y}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p4'/n'm'm$ 64.6.458	$\{E, C_{2z}, C_{4z}\mathcal{T}, C_{4z}^{-1}\mathcal{T}, C_{2y}, C_{2x}, C_{2xy}\mathcal{T}, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, \bar{C}_{4z}, \bar{C}_{4z}^{-1}, M_y\mathcal{T}, M_x\mathcal{T}, M_{xy}, M_{\bar{x}y}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/a$ 7.3.30	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2y}, \mathcal{PT}, M_y\mathcal{T}\}$	0	+
			$\mathrm{PL}_{(100)}$	$p2_1/m'11$ 15.3.78	$\{E, C_{2x}, \mathcal{PT}, M_x\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2xy}\mathcal{T}, \mathcal{PT}, M_{xy}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2\bar{x}y}\mathcal{T}, \mathcal{PT}, M_{\bar{x}y}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
78 66.3.466 $p^{-1}\bar{3}$ SI p. 80	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$	$\mathrm{AX}_{[001]}$	$p\bar{3}'$ 66.3.466	$\{E, C_{3z}, C_{3z}^{-1}, \mathcal{PT}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}\}$	0	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
79 71.3.481 $p^{-1}\bar{3}^{1}1^{-1}m$ SI p. 80	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[1\bar{1}0]}, \mathcal{P}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[120]}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[120]}, \mathcal{P}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{AX}^{*}_{[210]}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[210]}, \mathcal{P}\mathcal{T}, M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{3}'1m$ 71.5.483	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[210]}\mathcal{T}, C_{2,[120]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[120]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[120]}\}$	+	0
			$\mathrm{PL}^{*}_{(100)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[210]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[210]}\}$	+	+
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[1\bar{1}0]}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
80 71.5.483 $p^{-1}\bar{3}^{1}1^{1}m$ SI p. 81	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}^{*}_{[1\bar{1}0]}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[1\bar{1}0]}\}$	+	+
			$\mathrm{AX}_{[120]}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[120]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[120]}\}$	+	0
			$\mathrm{AX}^{*}_{[210]}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[210]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[210]}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{3}'1m'$ 71.3.481	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}, \mathcal{P}\mathcal{T}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_{[210]}\mathcal{T}, M_{[120]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[120]}, \mathcal{P}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{PL}^{*}_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[210]}, \mathcal{P}\mathcal{T}, M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{PL}^{*}_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[1\bar{1}0]}, \mathcal{P}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+
81 72.3.486 $p^{-1}\bar{3}^{-1}m^{1}1$ SI p. 81	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$	$\mathrm{AX}_{[100]}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[100]}, \mathcal{P}\mathcal{T}, M_{[100]}\mathcal{T}\}$	+	0
			$\mathrm{AX}^{*}_{[010]}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[010]}, \mathcal{P}\mathcal{T}, M_{[010]}\mathcal{T}\}$	+	+
			$\mathrm{AX}^{*}_{[110]}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[110]}, \mathcal{P}\mathcal{T}, M_{[110]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{3}'m1$ 72.5.488	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[110]}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[010]}\mathcal{T}, \mathcal{P}\mathcal{T}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_{[110]}, M_{[100]}, M_{[010]}\}$	0	0
			$\mathrm{PL}^{*}_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[110]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[110]}\}$	+	+
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[100]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[100]}\}$	0	+
			$\mathrm{PL}^{*}_{(1\bar{2}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[010]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[010]}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
82 72.5.488 $p^{-1}\bar{3}^1m^11$ SI p. 81	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$	$\mathrm{AX}_{[100]}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[100]}\mathcal{T}, \mathcal{PT}, M_{[100]}\}$	0	+
			$\mathrm{AX}^*_{[010]}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[010]}\mathcal{T}, \mathcal{PT}, M_{[010]}\}$	+	+
			$\mathrm{AX}^*_{[110]}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[110]}\mathcal{T}, \mathcal{PT}, M_{[110]}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p\bar{3}'m'1$ 72.3.486	$\{E, C_{3z}, C_{3z}^{-1}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}, \mathcal{PT}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_{[110]}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[010]}\mathcal{T}\}$	0	0
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[110]}, \mathcal{PT}, M_{[110]}\mathcal{T}\}$	+	+
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[100]}, \mathcal{PT}, M_{[100]}\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[010]}, \mathcal{PT}, M_{[010]}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
83 75.3.497 $p^16/^{-1}m$ SI p. 82	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$	$\mathrm{AX}_{[001]}$	$p6/m'$ 75.3.497	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}, C_{6z}^{-1}, C_{6z}, \mathcal{PT}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+
84 75.5.499 $p^{-1}6/^1m$ SI p. 82	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$	$\mathrm{AX}_{[001]}$	$p6'/m$ 75.5.499	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, \mathcal{PT}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_z, \bar{C}_{6z}^{-1}, \bar{C}_{6z}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
85 80.3.522 $p^16/^{-1}m^{-1}m^{-1}m$ SI p. 82	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[100]}, C_{2,[120]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{[100]}\mathcal{T}, M_{[120]}\}$	+	0
			$\mathrm{AX}^*_{[010]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[010]}, C_{2,[210]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{[010]}\mathcal{T}, M_{[210]}\}$	+	+
			$\mathrm{AX}^*_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[110]}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}, \mathcal{P}\mathcal{T}, M_z, M_{[110]}, M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[120]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[120]}, \mathcal{P}\mathcal{T}, M_z, M_{[100]}, M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{AX}^*_{[210]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[010]}\mathcal{T}, C_{2,[210]}, \mathcal{P}\mathcal{T}, M_z, M_{[010]}, M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p6/m'mm$ 80.5.524	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}, C_{6z}^{-1}, C_{6z}, C_{2,[110]}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}, C_{2,[120]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}, M_{[110]}, M_{[100]}, M_{[010]}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[120]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[120]}\}$	+	0
			$\mathrm{PL}^*_{(100)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[210]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[210]}\}$	+	+
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[110]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[110]}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[1\bar{1}0]}\}$	+	+
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[100]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[100]}\}$	0	+
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[010]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[010]}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
86 80.5.524 $p^16/^{-1}m^1m^1m$ SI p. 83	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{1\|C_{2z}\}$ $\{1\|C_{6z}^{-1}\}$ $\{1\|C_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{-1\|M_z\}$ $\{-1\|\bar{C}_{6z}^{-1}\}$ $\{-1\|\bar{C}_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[120]}, \mathcal{PT}, M_z, M_{[100]}, M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{AX}^*_{[010]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[010]}\mathcal{T}, C_{2,[210]}, \mathcal{PT}, M_z, M_{[010]}, M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[110]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}, \mathcal{PT}, M_z, M_{[110]}, M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{AX}^*_{[1\bar{1}0]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[110]}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{PT}, M_z, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\}$	+	+
			$\mathrm{AX}_{[120]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[100]}, C_{2,[120]}\mathcal{T}, \mathcal{PT}, M_z, M_{[100]}\mathcal{T}, M_{[120]}\}$	+	0
			$\mathrm{AX}^*_{[210]}$	$cmm'm$ 47.7.336	$\{E, C_{2z}\mathcal{T}, C_{2,[010]}, C_{2,[210]}\mathcal{T}, \mathcal{PT}, M_z, M_{[010]}\mathcal{T}, M_{[210]}\}$	+	+
			$\mathrm{AX}_{[001]}$	$p6/m'm'm'$ 80.3.522	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}, C_{6z}^{-1}, C_{6z}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}, \mathcal{PT}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_z\mathcal{T}, \bar{C}_{6z}^{-1}\mathcal{T}, \bar{C}_{6z}\mathcal{T}, M_{[110]}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[010]}\mathcal{T}, M_{[210]}\mathcal{T}, M_{[120]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p11'2'/m$ 6.3.23	$\{E, C_{2z}\mathcal{T}, \mathcal{PT}, M_z\}$	+	+
			$\mathrm{PL}_{(010)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[120]}, \mathcal{PT}, M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{PL}^*_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[210]}, \mathcal{PT}, M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[110]}, \mathcal{PT}, M_{[110]}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[1\bar{1}0]}, \mathcal{PT}, M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[100]}, \mathcal{PT}, M_{[100]}\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[010]}, \mathcal{PT}, M_{[010]}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
87 80.8.527 $p^{-1}6/{}^1m^1m^{-1}m$ SI p. 83	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{-1\|C_{2,[110]}\}$ $\{-1\|C_{2,[100]}\}$ $\{-1\|C_{2,[010]}\}$ $\{1\|C_{2,[210]}\}$ $\{1\|C_{2,[120]}\}$ $\{1\|C_{2,[1\bar{1}0]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{1\|M_{[110]}\}$ $\{1\|M_{[100]}\}$ $\{1\|M_{[010]}\}$ $\{-1\|M_{[210]}\}$ $\{-1\|M_{[120]}\}$ $\{-1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2,[100]}\mathcal{T}, C_{2,[120]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_{[100]}, M_{[120]}\}$	0	0
			$\mathrm{AX}_{[010]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_{[010]}, M_{[210]}\}$	0	0
			$\mathrm{AX}_{[110]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_{[110]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2,[110]}, C_{2,[1\bar{1}0]}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[120]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2,[100]}, C_{2,[120]}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_{[100]}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[210]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2,[010]}, C_{2,[210]}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}, M_{[010]}\mathcal{T}, M_{[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'/mm'm$ 80.9.528	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, C_{2,[110]}, C_{2,[100]}, C_{2,[010]}, C_{2,[210]}\mathcal{T}, C_{2,[120]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_z, \bar{C}_{6z}^{-1}, \bar{C}_{6z}, M_{[110]}\mathcal{T}, M_{[100]}\mathcal{T}, M_{[010]}\mathcal{T}, M_{[210]}, M_{[120]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{P}\mathcal{T}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[120]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[120]}\}$	+	0
			$\mathrm{PL}^*_{(100)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[210]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[210]}\}$	+	+
			$\mathrm{PL}^*_{(110)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[110]}, \mathcal{P}\mathcal{T}, M_{[110]}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{P}\mathcal{T}, M_{[1\bar{1}0]}\}$	+	+
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[100]}, \mathcal{P}\mathcal{T}, M_{[100]}\mathcal{T}\}$	+	0
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[010]}, \mathcal{P}\mathcal{T}, M_{[010]}\mathcal{T}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{P}\mathcal{T}\}$	+	+

cSLG label and symbol	Nonrelativistic class	Operation in SLG	Néel direction	SOC MLG symbol and number	Operation in MLG	$\chi^{t,b}_{xyy}$	$\chi^{t,b}_{yxx}$
88 80.9.528 $p^{-1}6/^1m^{-1}m^1m$ SI p. 84	PT AFM	$\{1\|E\}$ $\{1\|C_{3z}\}$ $\{1\|C_{3z}^{-1}\}$ $\{-1\|C_{2z}\}$ $\{-1\|C_{6z}^{-1}\}$ $\{-1\|C_{6z}\}$ $\{1\|C_{2,[110]}\}$ $\{1\|C_{2,[100]}\}$ $\{1\|C_{2,[010]}\}$ $\{-1\|C_{2,[210]}\}$ $\{-1\|C_{2,[120]}\}$ $\{-1\|C_{2,[1\bar{1}0]}\}$ $\{-1\|\mathcal{P}\}$ $\{-1\|\bar{C}_{3z}\}$ $\{-1\|\bar{C}_{3z}^{-1}\}$ $\{1\|M_z\}$ $\{1\|\bar{C}_{6z}^{-1}\}$ $\{1\|\bar{C}_{6z}\}$ $\{-1\|M_{[110]}\}$ $\{-1\|M_{[100]}\}$ $\{-1\|M_{[010]}\}$ $\{1\|M_{[210]}\}$ $\{1\|M_{[120]}\}$ $\{1\|M_{[1\bar{1}0]}\}$	$\mathrm{AX}_{[100]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2,[100]}, C_{2,[120]}, \mathcal{PT}, M_z\mathcal{T}, M_{[100]}\mathcal{T}, M_{[120]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[010]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2,[010]}, C_{2,[210]}, \mathcal{PT}, M_z\mathcal{T}, M_{[010]}\mathcal{T}, M_{[210]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[110]}$	$cm'm'm'$ 47.3.332	$\{E, C_{2z}, C_{2,[110]}, C_{2,[1\bar{1}0]}, \mathcal{PT}, M_z\mathcal{T}, M_{[110]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{AX}_{[1\bar{1}0]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2,[110]}\mathcal{T}, C_{2,[1\bar{1}0]}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_{[110]}, M_{[1\bar{1}0]}\}$	0	0
			$\mathrm{AX}_{[120]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2,[100]}\mathcal{T}, C_{2,[120]}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_{[100]}, M_{[120]}\}$	0	0
			$\mathrm{AX}_{[210]}$	$cmmm'$ 47.6.335	$\{E, C_{2z}, C_{2,[010]}\mathcal{T}, C_{2,[210]}\mathcal{T}, \mathcal{PT}, M_z\mathcal{T}, M_{[010]}, M_{[210]}\}$	0	0
			$\mathrm{AX}_{[001]}$	$p6'/mmm'$ 80.8.527	$\{E, C_{3z}, C_{3z}^{-1}, C_{2z}\mathcal{T}, C_{6z}^{-1}\mathcal{T}, C_{6z}\mathcal{T}, C_{2,[110]}\mathcal{T}, C_{2,[100]}\mathcal{T}, C_{2,[010]}\mathcal{T}, C_{2,[210]}, C_{2,[120]}, C_{2,[1\bar{1}0]}, \mathcal{PT}, \bar{C}_{3z}\mathcal{T}, \bar{C}_{3z}^{-1}\mathcal{T}, M_z, \bar{C}_{6z}^{-1}, \bar{C}_{6z}, M_{[110]}, M_{[100]}, M_{[010]}, M_{[210]}\mathcal{T}, M_{[120]}\mathcal{T}, M_{[1\bar{1}0]}\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(001)}$	$p112/m'$ 6.4.24	$\{E, C_{2z}, \mathcal{PT}, M_z\mathcal{T}\}$	0	0
			$\mathrm{PL}_{(010)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[120]}, \mathcal{PT}, M_{[120]}\mathcal{T}\}$	0	+
			$\mathrm{PL}^*_{(100)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[210]}, \mathcal{PT}, M_{[210]}\mathcal{T}\}$	+	+
			$\mathrm{PL}^*_{(110)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[110]}\mathcal{T}, \mathcal{PT}, M_{[110]}\}$	+	+
			$\mathrm{PL}^*_{(1\bar{1}0)}$	$c2/m'11$ 18.3.97	$\{E, C_{2,[1\bar{1}0]}, \mathcal{PT}, M_{[1\bar{1}0]}\mathcal{T}\}$	+	+
			$\mathrm{PL}_{(2\bar{1}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[100]}\mathcal{T}, \mathcal{PT}, M_{[100]}\}$	0	+
			$\mathrm{PL}^*_{(1\bar{2}0)}$	$c2'/m11$ 18.5.99	$\{E, C_{2,[010]}\mathcal{T}, \mathcal{PT}, M_{[010]}\}$	+	+
			GEN	$p\bar{1}'$ 2.3.6	$\{E, \mathcal{PT}\}$	+	+

Appendix A. Complete coupled-component constraints

$A^\ell = \chi^\ell_{xyy}$, $B^\ell = \chi^\ell_{yxx}$. Each row below gives the entire allowed linear space for a starred branch, with independent real parameters u or u_1, u_2. These relations supplement the $+, -, 0, v$ flags; they are not additional assumptions.

No.	cSLG label	Néel direction	Complete allowed space
62	61.3.424	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
62	61.3.424	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
62	61.3.424	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
62	61.3.424	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
63	61.5.426	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
63	61.5.426	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
63	61.5.426	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
63	61.5.426	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
64	61.6.427	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
64	61.6.427	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
64	61.6.427	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
64	61.6.427	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
65	61.7.428	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
65	61.7.428	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
65	61.7.428	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
65	61.7.428	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
66	62.3.437	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
66	62.3.437	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
66	62.3.437	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
66	62.3.437	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
67	62.5.439	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
67	62.5.439	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
67	62.5.439	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
67	62.5.439	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
68	62.6.440	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
68	62.6.440	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
68	62.6.440	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
68	62.6.440	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
69	62.7.441	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
69	62.7.441	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
69	62.7.441	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
69	62.7.441	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
70	63.3.446	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
70	63.3.446	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
70	63.3.446	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
70	63.3.446	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
71	63.5.448	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
71	63.5.448	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
71	63.5.448	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
71	63.5.448	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
72	63.6.449	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
72	63.6.449	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
72	63.6.449	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
72	63.6.449	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
73	63.7.450	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
73	63.7.450	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$

No.	cSLG label	Néel direction	Complete allowed space
73	63.7.450	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
73	63.7.450	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
74	64.3.455	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
74	64.3.455	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
74	64.3.455	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
74	64.3.455	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
75	64.5.457	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
75	64.5.457	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
75	64.5.457	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
75	64.5.457	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
76	64.6.458	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
76	64.6.458	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
76	64.6.458	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
76	64.6.458	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
77	64.7.459	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
77	64.7.459	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
77	64.7.459	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-1,\ 1,\ -1,\ 1\big)$
77	64.7.459	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(1,\ 1,\ 1,\ 1\big)$
79	71.3.481	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
79	71.3.481	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
79	71.3.481	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
79	71.3.481	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
80	71.5.483	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
80	71.5.483	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
80	71.5.483	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
80	71.5.483	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
81	72.3.486	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
81	72.3.486	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
81	72.3.486	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
81	72.3.486	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
82	72.5.488	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
82	72.5.488	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
82	72.5.488	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
82	72.5.488	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
85	80.3.522	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
85	80.3.522	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
85	80.3.522	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
85	80.3.522	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
85	80.3.522	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
85	80.3.522	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
85	80.3.522	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
85	80.3.522	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
86	80.5.524	$\mathrm{AX}_{[010]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
86	80.5.524	$\mathrm{AX}_{[110]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$
86	80.5.524	$\mathrm{AX}_{[1\bar{1}0]}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
86	80.5.524	$\mathrm{AX}_{[210]}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 3,\ -\sqrt{3},\ 3\big)$
86	80.5.524	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 1,\ \sqrt{3},\ 1\big)$
86	80.5.524	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\ 3,\ \sqrt{3},\ 3\big)$
86	80.5.524	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\ 1,\ -\sqrt{3},\ 1\big)$

No.	cSLG label	Néel direction	Complete allowed space
86	80.5.524	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, 3,\, -\sqrt{3},\, 3\big)$
87	80.8.527	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, 3,\, -\sqrt{3},\, 3\big)$
87	80.8.527	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, 3,\, \sqrt{3},\, 3\big)$
87	80.8.527	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, 3,\, \sqrt{3},\, 3\big)$
87	80.8.527	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, 3,\, -\sqrt{3},\, 3\big)$
88	80.9.528	$\mathrm{PL}_{(100)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, 1,\, \sqrt{3},\, 1\big)$
88	80.9.528	$\mathrm{PL}_{(110)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, 1,\, -\sqrt{3},\, 1\big)$
88	80.9.528	$\mathrm{PL}_{(1\bar{1}0)}$	$(A^t, B^t, A^b, B^b) = u\big(-\sqrt{3},\, 1,\, -\sqrt{3},\, 1\big)$
88	80.9.528	$\mathrm{PL}_{(1\bar{2}0)}$	$(A^t, B^t, A^b, B^b) = u\big(\sqrt{3},\, 1,\, \sqrt{3},\, 1\big)$

Appendix B. Layer relation enforced by PT

For the inversion point operation $R_{\mathcal{P}} = -I_3$, an axial Néel vector obeys $\det(R_{\mathcal{P}})R_{\mathcal{P}}\hat{\mathbf{n}} = \hat{\mathbf{n}}$. The spin-reversing operation $\{-1\|\mathcal{P}\}$ therefore gives the SOC symmetry $\mathcal{PT}$ for every collinear direction. Spatial inversion exchanges the physical layers and changes the sign of a polar third-rank tensor. Time reversal changes the sign of the intrinsic Hall tensor considered here. Their product has positive tensor parity:

$$\chi^{b}_{abc} = (-1)_{\mathcal{T}}(-1)^{3}_{\mathcal{P}}\chi^{t}_{abc} = \chi^{t}_{abc}.$$

Equivalently, the reduced Hall vector transforms with $qR_{\|} = (-1)(-I_2) = I_2$, so

$$(A^t, B^t, A^b, B^b) = (A, B, A, B).$$

Other retained symmetries restrict (A, B) further. An allowed component is marked $+$; if forbidden in each layer, it is marked 0. Additional relations between A and B are given in Appendix A. An allowed component contributes $\chi^{\mathrm{tot}} = 2\chi^{t}$ under the same layer normalization; PT does not enforce an opposite-sign hidden pair. The actual magnitude and any accidental zero require a microscopic calculation.

The flag distribution across the 625 direction branches is:

$(\chi^{t,b}_{xyy}, \chi^{t,b}_{yxx})$ flags	Branches
$(+,+)$	243
$(+,0)$	136
$(0,+)$	118
$(0,0)$	128

These counts refer to source-coordinate direction strata, not distinct MLG types or numbers of materials. In the 104 starred branches, two allowed components need not be independent.